

Product Backlog for Agentic Lender

Year 3 Group Project

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Product Vision

Agentic Lender is a demo mortgage application system. A borrower applies through a mobile app, and an AI committee automatically checks whether they can afford the loan and whether the property is fairly priced a human broker or a bank clerk reviews the AI's recommendation on a web dashboard and makes the final call and once approved, the loan is permanently recorded so the decision can't be altered later. Our goal is to show that a lot of the slow and tedious manual work in mortgage approval can be sped up with automation while still keeping employees in charge of the final call

Backlog Legend

Task: A task is an individual piece of work that will be done along with other tasks to complete an epic.

Priority: (Must/ Should/ Could) is what's essential, what we should have and what's nice to have.

Hours: Rough estimate of how many hours a task will take to complete.

Sprint: Which of the three 3 week sprints this task is targeted for.

AL Mobile App for Applicants

Epic 1: Authentication & Onboarding

Let borrowers securely create an account and sign in.

Task	Priority	Hours	Sprint
Build a register/login screen that requires an	Must	3	1

email and password.			
Integrate Firebase Authentication.	Must	5	1

Epic 2: Affordability & loan Application

Capture the borrower's financial details to start an application.

Task	Priority	Hours	Sprint
Build an application form that requests income, debts, deposit, term, etc.	must	4	1
Validate the forms inputs and submit to backend API.	must	5	1

Epic 3: Fair price property checker

Let borrowers check if a property is fairly priced.

Task	Priority	Hours	Sprint
Build Property spec input screen that takes location, size, bedrooms, bathrooms, square foot, etc.	Must	6	2
Call the ML valuation API and display the results.	Must	3	2

Epic 4: AI Status & Loan Tracking

Show the borrower where their application stands.

Task	Priority	Hours	Sprint
Build approval likelihood screen.	Must	3	2

Build loan lifecycle status tracker UI.	Should	5	2
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Epic 5: Broker Communication

Let borrowers talk directly to their assigned broker/official.

Task	Priority	Hours	Sprint
Build a messaging screen.	Should	8	3

Epic 6: Payments

Let borrowers pay to fast track appraisal/verification.

Task	Priority	Hours	Sprint
Integrate Stripe/PayPal sandbox for appraisal fee payment.	Should	4	3
Add some sort of tag to appraisal request to let brokers know it should join a priority queue.	Should	2	3

Epic 7: Support & Reviews

Let borrowers get help and leave feedback.

Task	Priority	Hours	Sprint
Build support request form .	Could	3.5	3
Build rating/review submission screen.	Could	3.5	3

AL Broker and Underwriter Portal

Epic 8: Authentication & Role Based Access

Let brokers and underwriters sign in with the right permissions.

Task	Priority	Hours	Sprint
Build a login screen with role detection.	Must	4.5	1
Implement broker vs underwriter permission logic.	Must	6	1

Epic 9: Origination Pipeline View

Give staff a queue of applications to work through.

Task	Priority	Hours	Sprint
Build pipeline queue view and general information and status of loans.	Must	6	2
Add filtering and sorting to the pipeline view.	Should	5	2

Epic 10: Loan File & Collateral Assessment

Show staff the full detail behind the one application.

Task	Priority	Hours	Sprint
Build loan detail view, asking price vs valuation and debt to income ratio (this is more detailed than the general information task above).	Must	6	2

Epic 11: AI Audit & Deliberation Console

Make the AI committees reasoning visible and audible.

Task	Priority	Hours	Sprint
Setup access to the agent council.	Must	3.5	2
Build console displaying agent reasoning logs and consensus.	Must	3	2

Epic 12: Macro Stress Testing Simulator

Let managers test the portfolio against economic shocks.

Task	Priority	Hours	Sprint
Research and choose available API's to test portfolio against.	Should	3	3
Build interactive rate shock/ downturn simulator UI.	Should	7	3

Epic 13: On Chain Settlement & Export

Let underwriters finalise approvals and produce audit records.

Task	Priority	Hours	Sprint
Build sign off action that triggers smart contract loan minting	Must	7.5	3
Build PDF/CSV audit pack export.	Should	5	3

AL Core, Database and Backend

Epic 14: Data Store Setup

Give every subsystem a shared consistent source of data.

Task	Priority	Hours	Sprint
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Design firebase schema (users, document metadata, broker, assignments, application state).	Must	6	1
Wire firebase authentication across mobile and web.	Must	5	1

Execution Environment

Stand up somewhere to deploy and test smart contracts.

Task	Priority	Hours	Sprint
Set up Hardhat/Anvil smart contract via Docker.	Must	6	1

Loan Tokenisation Contract

Represent an approved loan as an on chain digital record.

Task	Priority	Hours	Sprint
Research ERC-721 and similar tokenisers and chose a standard.	Must	3	2
Write and test loan tokenisation smart contract ERC-721 or equivalent	Must	8	2

Immutable Audit Registry

Make approval decisions tamper proof.

Task	Priority	Hours	Sprint
Commit SHA 256 hash of credit memo, valuation	Must	8	3

data and approval on chain.			
Sync Firestore application state with on chain lifecycle state.	Should	6	3

AL AI Engine Predictive ML and Multi Agent Committee

AI Service Infrastructure

Give Mobile and Web a service to talk to for ML/AI features.

Task	Priority	Hours	Sprint
Set up Flask/FastAPI.	Must	3	1
Create service linking mobile and web to AI layer with Flask/FastAPI.	Must	5	1

Property Valuation Model

Predict a fair price for a property

Task	Priority	Hours	Sprint
Source and clean a housing dataset.	Must	5	1
Train and evaluate a regression valuation model.	Must	8	1

Affordability Agent

Check whether the borrower can actually afford the loan.

Task	Priority	Hours	Sprint
Build agent evaluating disposable income, stress tested repayments and	Must	8	2

regulatory borrowing caps.			
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Valuation Agent

Check whether the Property is fairly priced.

Task	Priority	Hours	Sprint
Build agent comparing asking price to ML valuation and computing LTV.	Must	8	2

Market and Macro Researcher Agent

Bring in live interest rate context.

Task	Priority	Hours	Sprint
Research and choose a web search tool to get current mortgage rates and trends from.	Should	2	3
Build agent pulling current base rates/mortgage trends via chosen web search tool.	Should	6	3

Underwriting Strategist Agent

Turn all agent outputs into one final decision.

Task	Priority	Hours	Sprint
Build agent that deliberates on other agents outputs and issues a decision and credit memo	Must	8	3

Multi LLM Integration

Diversify Council with cloud and locally hosted LLM's

Task	Priority	Hours	Sprint
Integrate 2 cloud LLMS and 1 locally hosted LLM via Ollama into agent consensus.	Must	9	3

Custom Feature

Custom Feature (not decided yet)

Implement at least one custom feature **Choose before Sprint 3 submission**

Task	Priority	Hours	Sprint
Option A Document OCR Pipeline: Android camera or Web document upload module using Tesseract/OCR to parse pay slips and extract applicant income and deductions automatically.	Could	12	3
Option B Green Mortgage Smart Incentive: Smart contract logic that inspects a property's Building Energy Rating (BER/EPC) and automatically deducts 25 bps	Could	12	3

from the contract's baseline APR for energy-efficient homes.			
Option C First-Time Buyer Explainer Bot: Conversational in-app assistant fine-tuned on regional mortgage terminology, government first-home schemes, and loan closing costs.	Coul	12	3
Option D Neighbourhood Amenity Scoring Agent: Agentic integration with mapping APIs (Google Maps / OpenStreetMap) to score property proximity to schools, transit, and amenities as a risk modifier.	Could	12	3